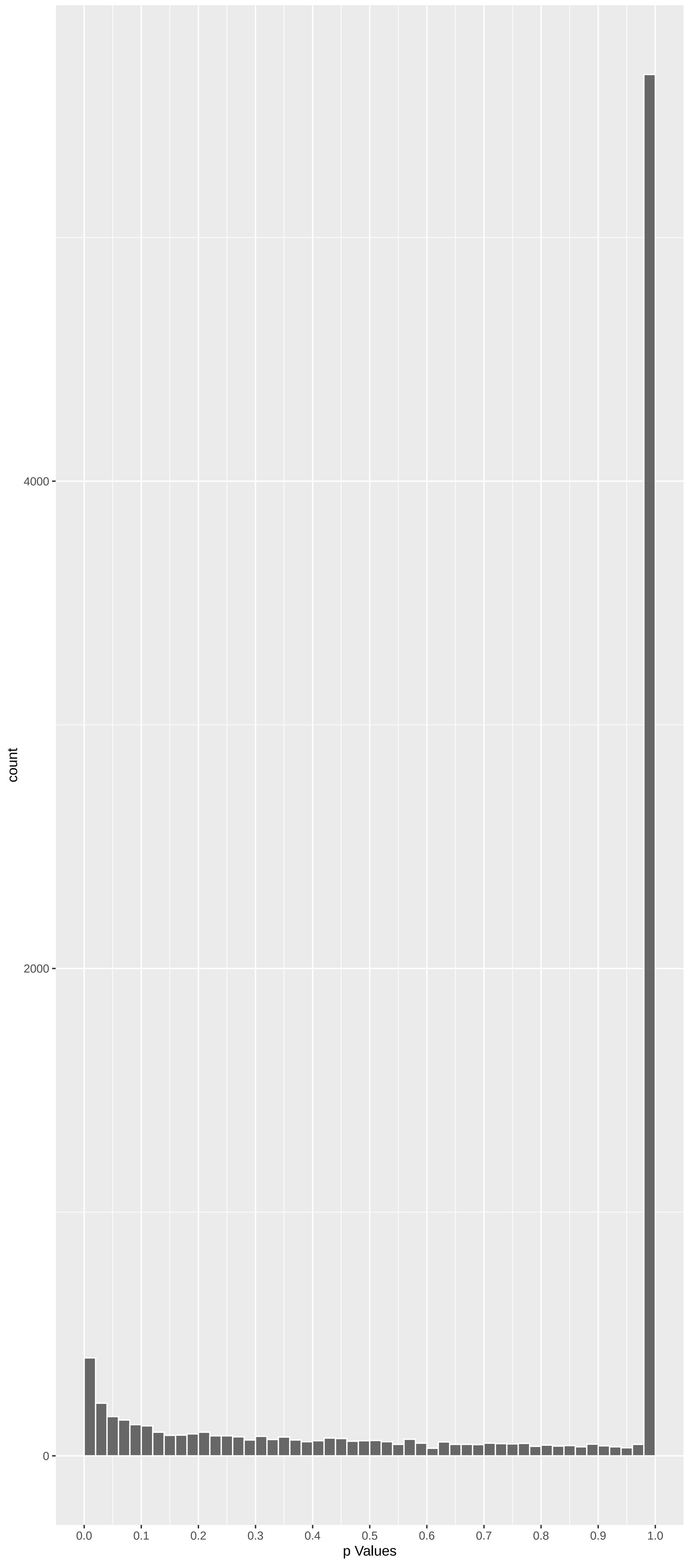
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
CLPS	5.007393	3.310332e−06	4.214e−02	1.000e+00
BPIFB1	4.868107	6.760323e−06	4.303e−02	1.000e+00
MON1A	4.727082	1.366614e−05	5.799e−02	1.000e+00
CYP11A1	4.645477	2.035754e−05	6.385e−02	1.000e+00
TASOR	4.569636	2.931433e−05	6.385e−02	1.000e+00
HMCES	4.564129	3.009437e−05	6.385e−02	1.000e+00
ACRBP	4.490017	4.273056e−05	6.746e−02	1.000e+00
RBSN	4.338096	8.623352e−05	6.746e−02	1.000e+00
SLC1A1	4.371120	7.416665e−05	6.746e−02	1.000e+00
PARP12	4.318223	9.437424e−05	6.746e−02	1.000e+00
IL1R2	4.317289	9.477457e−05	6.746e−02	1.000e+00
SLC23A1	4.379575	7.134645e−05	6.746e−02	1.000e+00
TRIM67	4.456170	5.006214e−05	6.746e−02	1.000e+00
RNF216	4.345619	8.333009e−05	6.746e−02	1.000e+00
FAR2	4.315856	9.539137e−05	6.746e−02	1.000e+00
FGD5	4.332901	8.829456e−05	6.746e−02	1.000e+00
SH3D19	4.401181	6.459798e−05	6.746e−02	1.000e+00
LYPD4	4.438037	5.446973e−05	6.746e−02	1.000e+00
OTUD6B	4.243684	1.319279e−04	8.839e−02	1.000e+00
ENTPD5	4.230074	1.401688e−04	8.922e−02	1.000e+00
IFNK	4.180398	1.745997e−04	1.010e−01	1.000e+00
COPRS	4.180419	1.745834e−04	1.010e−01	1.000e+00
ZNF446	4.164049	1.875921e−04	1.038e−01	1.000e+00
GPRC5A	4.114402	2.329107e−04	1.235e−01	1.000e+00
PIP5K1A	4.104536	2.430761e−04	1.238e−01	1.000e+00

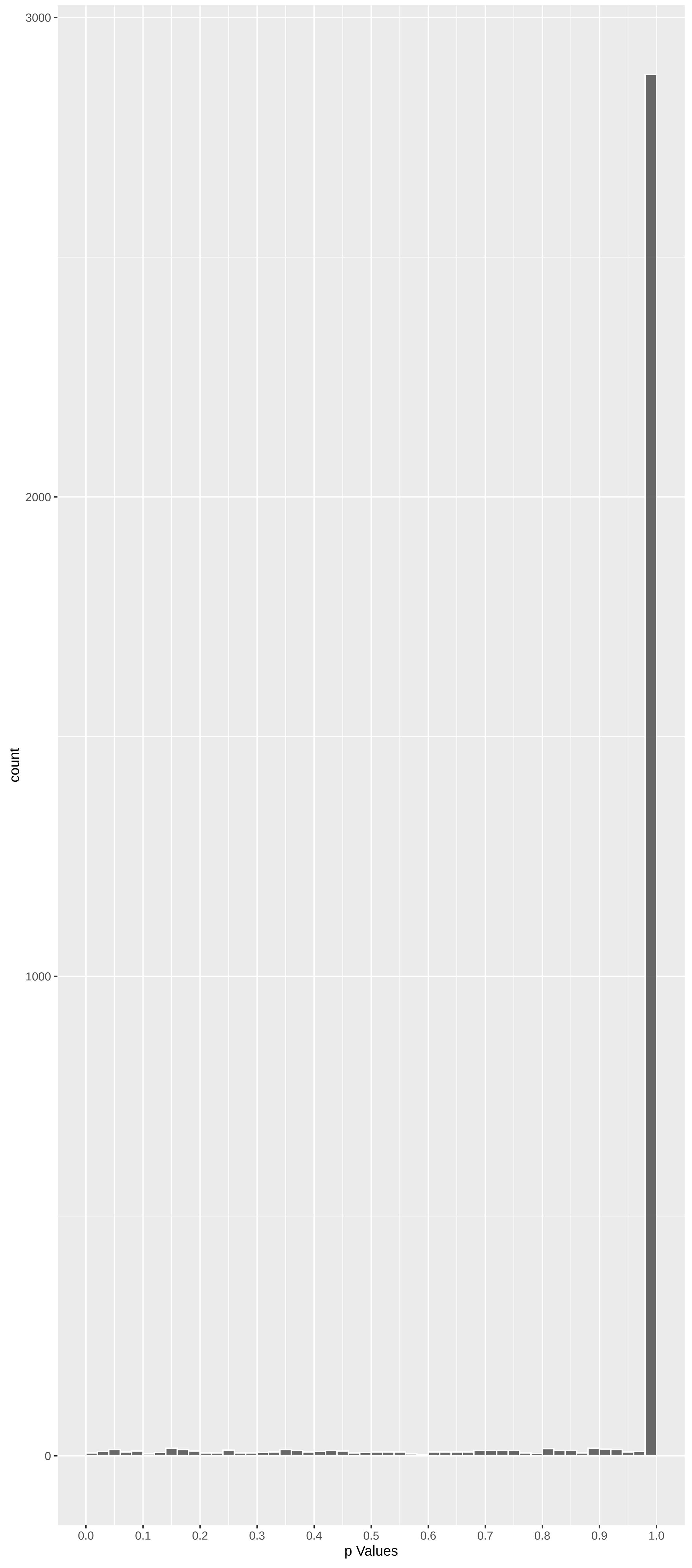
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	0.94674959	1.00000000	1.000e+00	1.000e+00
FKBP4	2.34242783	0.11495243	1.000e+00	1.000e+00
NDUFAF7	0.48308779	1.00000000	1.000e+00	1.000e+00
FUCA2	−0.02802245	1.00000000	1.000e+00	1.000e+00
DBNDD1	0.16241524	1.00000000	1.000e+00	1.000e+00
HS3ST1	1.32646839	1.00000000	1.000e+00	1.000e+00
SEMA3F	1.94391382	0.31143506	1.000e+00	1.000e+00
CFTR	0.32807344	1.00000000	1.000e+00	1.000e+00
CYP51A1	2.21907984	0.15888776	1.000e+00	1.000e+00
USP28	−0.49015140	1.00000000	1.000e+00	1.000e+00
SLC7A2	0.18054877	1.00000000	1.000e+00	1.000e+00
PDK4	1.88873166	0.35356677	1.000e+00	1.000e+00
USH1C	−0.33569601	1.00000000	1.000e+00	1.000e+00
RALA	1.45351244	0.87648917	1.000e+00	1.000e+00
BAIAP2L1	3.09433247	0.01183538	4.959e−01	1.000e+00
CACNG3	0.43199583	1.00000000	1.000e+00	1.000e+00
TMEM132A	0.08227171	1.00000000	1.000e+00	1.000e+00
DVL2	−1.93076116	0.32107558	1.000e+00	1.000e+00
RPAP3	0.47227357	1.00000000	1.000e+00	1.000e+00
TRAPPC6A	−0.21792823	1.00000000	1.000e+00	1.000e+00
SPATA20	0.04211740	1.00000000	1.000e+00	1.000e+00
CD79B	0.28663882	1.00000000	1.000e+00	1.000e+00
RHBDD2	3.01417074	0.01546096	5.677e−01	1.000e+00
RPUSD1	2.18691437	0.17249266	1.000e+00	1.000e+00
TSR3	0.36822576	1.00000000	1.000e+00	1.000e+00

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
CLPS	5.007393	3.310332e−06	4.214e−02	1.000e+00
BPIFB1	4.868107	6.760323e−06	4.303e−02	1.000e+00
MON1A	4.727082	1.366614e−05	5.799e−02	1.000e+00
CYP11A1	4.645477	2.035754e−05	6.385e−02	1.000e+00
TASOR	4.569636	2.931433e−05	6.385e−02	1.000e+00
HMCES	4.564129	3.009437e−05	6.385e−02	1.000e+00
ACRBP	4.490017	4.273056e−05	6.746e−02	1.000e+00
RBSN	4.338096	8.623352e−05	6.746e−02	1.000e+00
SLC1A1	4.371120	7.416665e−05	6.746e−02	1.000e+00
PARP12	4.318223	9.437424e−05	6.746e−02	1.000e+00

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
MBNL2	−3.459932	0.003241874	3.328e−01	1.000e+00
CCL1	−3.348578	0.004873649	3.902e−01	1.000e+00
PGBD1	−3.226551	0.007517513	4.154e−01	1.000e+00
PLPP1	−3.029434	0.014700758	5.544e−01	1.000e+00
PTEN	−2.961736	0.018354593	6.025e−01	1.000e+00
PIR	−2.954781	0.018773492	6.064e−01	1.000e+00
CNN3	−2.929371	0.020378909	6.286e−01	1.000e+00
CAMK2B	−2.917131	0.021196054	6.409e−01	1.000e+00
GPR179	−2.848838	0.026327502	6.905e−01	1.000e+00
TRIM15	−2.808715	0.029843800	7.196e−01	1.000e+00

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.08675135	235	5.324e−06	1.614e−02	ANO6:95 SLC6A12:119 SLC36A2:138 HPN:160 SLC6A15:194 ATP6V0A4:196
Monoatomic Cation Transmembrane Transpor	0.08536139	234	7.845e−06	1.614e−02	ANO6:95 SLC6A12:119 SLC36A2:138 HPN:160 SLC6A15:194 ATP6V0A4:196
Monocarboxylic Acid Transport (GO:001571	0.18171165	50	9.015e−06	1.614e−02	SLC5A8:64 SLC5A12:67 SLC6A12:119 ABCD3:167 FABP1:179 KDM5A:401
Organic Hydroxy Compound Transport (GO:0	0.20888495	36	1.468e−05	1.971e−02	SLC22A1:62 SLC5A8:64 ABCD3:167 GHRL:252 SLC51B:455 SLC51A:698
Carboxylic Acid Transport (GO:0046942)	0.16184370	47	1.259e−04	1.353e−01	SLC5A12:67 SLC6A12:119 SLC36A2:138 SLC6A15:194 SLC6A18:429 SLC6A6:478
Intracellular Monoatomic Ion Homeostasis	0.23354165	19	4.269e−04	3.822e−01	CLDN16:103 CNNM4:204 SLC4A1:312 LETM1:510 SLC39A10:812 RHCG:813
Organic Anion Transport (GO:0015711)	0.11847502	72	5.213e−04	4.000e−01	SLCO5A1:111 SLC52A3:143 SLC4A9:176 SLC4A1:312 SLC4A3:390 MGST1:402
Sodium Ion Transmembrane Transport (GO:0	0.11890770	67	7.794e−04	5.232e−01	SLC6A12:119 SLC6A15:194 SLC9A4:319 SLC12A3:356 SCNN1G:358 SCNN1B:406
Organic Cation Transport (GO:0015695)	0.17407857	30	9.747e−04	5.817e−01	SLC22A1:62 SLC6A12:119 SLC18B1:126 SLC22A4:675 SLC5A7:766 SLC44A5:769
Bile Acid And Bile Salt Transport (GO:00	0.23220741	16	1.305e−03	5.843e−01	ABCD3:167 SLC51B:455 SLC51A:698 SLC10A2:869 SLC10A3:891 SLC27A5:1144
Nitrogen Compound Transport (GO:0071705)	0.08104306	133	1.301e−03	5.843e−01	SLC22A1:62 SLC5A8:64 SLC6A12:119 SLC36A2:138 SLC52A3:143 SLC6A15:194
Sulfur Compound Transport (GO:0072348)	0.20871940	20	1.238e−03	5.843e−01	SLC22A1:62 SLC26A4:296 SLC26A8:768 SLC13A1:798 SLC35B3:826 SLC25A10:848
L−phenylalanine Catabolic Process (GO:00	0.42004141	4	3.621e−03	6.145e−01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA
L−phenylalanine Metabolic Process (GO:00	0.42004141	4	3.621e−03	6.145e−01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA
Acid Secretion (GO:0046717)	0.31814180	7	3.561e−03	6.145e−01	DRD3:243 GHRL:252 SLC51B:455 SLC51A:698 SLC26A6:3075 SLC9A3R1:3295
Alpha−Amino Acid Biosynthetic Process (G	0.27832458	10	2.310e−03	6.145e−01	BCAT2:28 SDSL:134 CAD:730 SDS:847 CPS1:1137 BCAT1:2925
Canonical Wnt Signaling Pathway (GO:0060	−0.15408568	33	2.206e−03	6.145e−01	PTEN:5 WNT4:25 DVL2:140 FRAT1:358 FZD6:407 BCL9L:411
Erythrose 4−Phosphate/Phosphoenolpyruvat	0.42004141	4	3.621e−03	6.145e−01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA
Ethanol Metabolic Process (GO:0006067)	0.34505051	6	3.425e−03	6.145e−01	ACSS2:412 ADH4:1182 SULT1A2:1487 SULT1E1:2135 SULT1A3:2950 SULT1B1:3463
Gamma−Aminobutyric Acid Transport (GO:00	0.38776891	5	2.676e−03	6.145e−01	SLC6A12:119 SLC6A6:478 SLC6A11:646 SLC6A13:2492 SLC9A3R1:3295 NA
Innate Immune Response In Mucosa (GO:000	0.42198546	4	3.468e−03	6.145e−01	BPIFB1:2 NOS2:197 PLA2G1B:225 IL4:3480 NA NA
Inorganic Cation Import Across Plasma Me	0.09640985	85	2.172e−03	6.145e−01	SLC12A9:271 CLU:299 SLC12A6:305 SLC9A4:319 SLC12A3:356 SCNN1G:358
Monoatomic Cation Homeostasis (GO:005508	0.11232214	61	2.450e−03	6.145e−01	CLDN16:103 CNNM4:204 SLC12A9:271 SLC12A6:305 SLC12A3:356 SCNN1G:358
Mucosal Immune Response (GO:0002385)	0.28314470	10	1.935e−03	6.145e−01	BPIFB1:2 NOS2:197 PLA2G1B:225 FFAR2:426 IFNLR1:1338 RAB17:2714
Negative Regulation Of Gene Expression V	0.37637612	5	3.562e−03	6.145e−01	ZNF445:338 UHRF1:1311 USP7:1572 HELLS:1669 UHRF2:2788 NA
Negative Regulation Of miRNA Metabolic P	0.21677282	15	3.661e−03	6.145e−01	POU5F1:270 PDGFB:603 ESR1:1493 HIF1A:1560 NFATC4:1662 NFATC3:1690
Negative Regulation Of miRNA Transcripti	0.24167076	14	1.748e−03	6.145e−01	POU5F1:270 PDGFB:603 ESR1:1493 HIF1A:1560 NFATC4:1662 NFATC3:1690
Negative Regulation Of Single Stranded V	0.34856061	6	3.111e−03	6.145e−01	TASOR:5 MORC2:30 INPP5K:534 SETDB1:1016 MPHOSPH8:1841 HMGA2:8031.5
Positive Regulation Of DNA Methylation−D	0.31902278	8	1.782e−03	6.145e−01	TASOR:5 MORC2:30 SETDB1:1016 TRIM28:1724 MPHOSPH8:1841 PPHLN1:2236
Positive Regulation Of Heterochromatin F	0.26843169	10	3.294e−03	6.145e−01	TASOR:5 MORC2:30 SETDB1:1016 TRIM28:1724 MPHOSPH8:1841 PPHLN1:2236
Regulation Of Cell Adhesion (GO:0030155)	−0.08285646	105	3.441e−03	6.145e−01	TGM2:44 P4HB:70 RAC2:161 CXCR3:394 FUT2:400 FES:402
Sodium Ion Transport (GO:0006814)	0.10041948	74	2.872e−03	6.145e−01	SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC6A15:194 SLC12A3:356 SCNN1G:358
One−Carbon Compound Transport (GO:001975	0.15367119	29	4.205e−03	6.643e−01	AQP11:35 SLC4A9:176 SLC4A1:312 SLC4A3:390 RHAG:855 CA2:947
Sulfate Transport (GO:0008272)	0.26202506	10	4.121e−03	6.643e−01	SLC26A4:296 SLC26A8:768 SLC13A1:798 SLC25A10:848 SLC26A2:2397 SLC13A4:2561
Chromosome Condensation (GO:0030261)	0.18648092	19	4.909e−03	7.202e−01	H1−6:139 AKAP8L:527 PLK1:811 AIFM1:853 ERN2:964 NCAPH:1141
Fat−Soluble Vitamin Metabolic Process (G	0.20373483	16	4.793e−03	7.202e−01	CYP11A1:4 LRAT:112 NQO1:374 KDM5A:401 GGCX:505 CYP1A1:695
Intracellular pH Elevation (GO:0051454)	0.30573074	7	5.095e−03	7.202e−01	NOX1:213 TMEM175:657 CLN3:1383 SLC26A3:1580 OCA2:2165 SLC26A6:3075
Replication Fork Processing (GO:0031297)	−0.13348788	37	4.995e−03	7.202e−01	BRCA1:247 RAD51:284 MUS81:369 PARP1:4699.5 ATRX:4699.5 WRN:4699.5
Aromatic Amino Acid Family Catabolic Pro	0.26303706	9	6.292e−03	8.242e−01	TAT:546 GSTZ1:673 HPD:760 HGD:1024 IL4I1:1697 KMO:2841
Sodium Ion Homeostasis (GO:0055078)	0.19151404	17	6.278e−03	8.242e−01	SLC12A3:356 SCNN1G:358 SCNN1B:406 ATP1B3:477 SCNN1A:653 ATP4B:1118

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.12279528	219	4.404e−10	2.852e−06	SLC1A1:12 SLC22A1:62 SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC36A2:138
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.06514096	566	1.566e−07	3.382e−04	SLC1A1:12 AQP11:35 MTTP:36 ALB:51 SLC22A1:62 SLC5A8:64
WP_NRF2_PATHWAY	0.16050922	92	1.077e−07	3.382e−04	TXNRD3:58 SLC5A8:64 SLC5A12:67 MGST3:80 SLC2A9:173 SLC6A15:194
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.14702944	95	7.612e−07	1.232e−03	SLC1A1:12 SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC36A2:138 SLC4A9:176
WP_NUCLEAR_RECEPTORS_METAPATHWAY	0.09475113	222	1.257e−06	1.628e−03	EHHADH:29 ABCB11:56 TXNRD3:58 SLC5A8:64 SLC5A12:67 MGST3:80
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.15728318	77	1.880e−06	1.951e−03	SLC22A1:62 SLC6A12:119 SLC6A15:194 SLC13A2:205 SLC6A18:429 SLC6A6:478
REACTOME_SLC_TRANSPORTER_DISORDERS	0.14901738	85	2.109e−06	1.951e−03	SLC1A1:12 SLC36A2:138 NUP133:147 SLC2A9:173 SLC26A4:296 SLC12A6:305
CAIRO_HEPATOBLASTOMA_DN	0.09447100	196	5.490e−06	4.444e−03	SLC1A1:12 GYS2:34 ABCB11:56 SLC22A1:62 GCDH:78 HP:90
WP_TRANSSULFURATION_ONECARBON_METABOLISM	0.18126636	51	7.645e−06	5.501e−03	BCAT2:28 CHDH:127 GNMT:174 PHGDH:301 GPX4:371 TYMS:468
FEVR_CTNNB1_TARGETS_UP	0.05261648	502	6.327e−05	3.854e−02	CLPS:1 SLC1A1:12 ENTPD5:20 SPTA1:33 AQP11:35 MTTP:36
LEE_LIVER_CANCER_DENA_DN	0.14481559	63	7.142e−05	3.854e−02	EHHADH:29 PNLIP:37 GCDH:78 IL1RAP:97 CYP7A1:131 ABCD3:167
HSIAO_LIVER_SPECIFIC_GENES	0.08571511	183	6.695e−05	3.854e−02	GYS2:34 MTTP:36 ALB:51 LGALS4:60 TM4SF4:86 HP:90
REACTOME_BIOLOGICAL_OXIDATIONS	0.10142111	123	1.058e−04	4.952e−02	CYP11A1:4 PODXL2:66 MGST3:80 CYP7A1:131 COMT:180 CYP2F1:253
REACTOME_METABOLISM_OF_LIPIDS	0.04752273	581	1.071e−04	4.952e−02	CYP11A1:4 FAR2:18 PIP5K1A:25 EHHADH:29 MORC2:30 ALB:51
REACTOME_SIGNALING_BY_WNT	−0.08640085	161	1.614e−04	6.967e−02	WNT4:25 DVL2:140 RAC2:161 H3C4:239 VPS26A:283 FRAT1:358
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.19410420	30	2.348e−04	9.504e−02	SLC6A12:119 SLC36A2:138 SLC6A15:194 SLC6A18:429 SLC6A6:478 SLC43A1:553
WP_2Q13_COPY_NUMBER_VARIATION_SYNDROME	0.16072059	43	2.680e−04	1.021e−01	MERTK:70 IL1RL2:87 IL1RAP:97 ZC3H6:245 TTL:373 PSD4:380
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20544983	26	2.889e−04	1.040e−01	CYP11A1:4 CYP7A1:131 COMT:180 HSD17B3:574 HSD17B8:679 SULT2B1:692
WP_OXIDATIVE_STRESS_RESPONSE	0.20027966	27	3.170e−04	1.081e−01	NOX1:213 TXNRD1:282 NQO1:374 MGST1:402 CYP1A1:695 HMOX1:923
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.10684806	92	4.049e−04	1.311e−01	CYP11A1:4 MGST3:80 CYP7A1:131 COMT:180 CYP2F1:253 GPX4:371
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	0.06908213	210	5.853e−04	1.805e−01	SLC23A1:11 EHHADH:29 SLC6A12:119 MMUT:175 AGXT:219 IVD:264
REACTOME_DIGESTION_OF_DIETARY_LIPID	0.48407682	4	7.991e−04	2.070e−01	CLPS:1 PNLIP:37 PNLIPRP1:336 PNLIPRP2:442 NA NA
BLALOCK_ALZHEIMERS_DISEASE_DN	−0.03617836	765	7.857e−04	2.070e−01	PTEN:5 ATP6V1C1:10 STXBP1:64 RALB:67 PCDH11X:69 NDUFA8:88
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	0.08421263	136	7.164e−04	2.070e−01	EHHADH:29 GYS2:34 MTTP:36 ABCB11:56 SLC22A1:62 GCDH:78
WOO_LIVER_CANCER_RECURRENCE_DN	0.12599558	60	7.449e−04	2.070e−01	MTTP:36 GCDH:78 SLC2A9:173 AASS:288 ACADSB:561 SARDH:634
REACTOME_DISORDERS_OF_TRANSMEMBRANE_TRAN	0.08259665	133	1.031e−03	2.568e−01	SLC1A1:12 ABCB11:56 SLC36A2:138 NUP133:147 SLC2A9:173 SLC26A4:296
STARK_HYPPOCAMPUS_22Q11_DELETION_DN	0.22904477	17	1.079e−03	2.588e−01	HIRA:61 TRMT2A:169 COMT:180 ARVCF:608 MRPL40:735 ZDHHC8:933
REACTOME_HEME_DEGRADATION	0.29579473	10	1.200e−03	2.658e−01	ALB:51 FABP1:179 HMOX1:923 UGT1A1:925 SLCO1B1:1202 BLVRB:1531
REACTOME_BETA_CATENIN_INDEPENDENT_WNT_SI	−0.10463927	80	1.231e−03	2.658e−01	WNT4:25 DVL2:140 RAC2:161 FZD6:407 AGO3:414 RAC1:451
REACTOME_SENSORY_PERCEPTION_OF_TASTE	0.17407477	29	1.181e−03	2.658e−01	TAS2R41:71 TAS2R4:208 SCNN1G:358 TAS1R1:385 SCNN1B:406 TAS1R3:521
WP_FOCAL_ADHESION	−0.07403851	158	1.359e−03	2.825e−01	PTEN:5 BIRC3:58 CRK:158 RAC2:161 MAPK8:211 VEGFD:282
BERTUCCI_MEDULLARY_VS_DUCTAL_BREAST_CANC	−0.08427575	121	1.396e−03	2.825e−01	GUCY1A2:127 UACA:170 SNRPN:218 ADAMTS12:267 SPOCK1:286 ACTN1:4699.5
ANDERSEN_LIVER_CANCER_KRT19_DN	0.12370233	54	1.678e−03	2.841e−01	GNMT:174 PCYOX1:287 SELENBP1:414 PTGR1:540 HPD:760 AOX1:779
CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL	0.03351551	785	1.642e−03	2.841e−01	CYP11A1:4 HMCES:6 SLC1A1:12 EHHADH:29 GYS2:34 ABCB11:56
KEGG_TASTE_TRANSDUCTION	0.17611416	27	1.543e−03	2.841e−01	TAS2R41:71 TAS2R4:208 SCNN1G:358 TAS1R1:385 SCNN1B:406 ADCY4:460
REACTOME_TRANSLATION	−0.06703756	189	1.535e−03	2.841e−01	RPS8:32 MRPL39:78 SEC61B:106 MRPL37:139 FARSA:194 DAP3:213
REACTOME_DIGESTION_AND_ABSORPTION	0.23391898	15	1.711e−03	2.841e−01	CLPS:1 PNLIP:37 PNLIPRP1:336 PNLIPRP2:442 GUCA2A:558 SLC2A2:899
YOSHIMURA_MAPK8_TARGETS_UP	0.03201275	877	1.546e−03	2.841e−01	CLPS:1 SLC1A1:12 GYS2:34 PNLIP:37 RBP2:57 LGALS4:60
KEGG_FOCAL_ADHESION	−0.07267361	158	1.666e−03	2.841e−01	PTEN:5 BIRC3:58 CRK:158 RAC2:161 MAPK8:211 VEGFD:282
VERNOCHET_ADIPOGENESIS	0.24823034	13	1.944e−03	2.861e−01	HP:90 APCDD1:157 MC2R:224 CYP2F1:253 FFAR2:426 RARRES2:1197

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Kidney Calculi	0.17864677	51	1.042e−05	1.018e−01	PTH:77 CLDN16:103 SLC2A9:173 SLC13A2:205 AGXT:219 SLC4A1:312
Cystic Fibrosis	0.06158866	374	5.133e−05	1.671e−01	BPIFB1:2 SGSM3:46 ALB:51 NPC1:116 MYD88:125 NR1I2:133
Hyperkalemia	0.30300147	15	4.865e−05	1.671e−01	CYP11A1:4 SCNN1G:358 SCNN1B:406 SCNN1A:653 CA12:836 WNK1:975
Hypertensive disease	0.03945789	918	7.795e−05	1.903e−01	CYP11A1:4 ALB:51 TXNRD3:58 PTH:77 HP:90 PRKCD:91
Anemia, hereditary spherocytic hemolytic	0.39463988	5	2.243e−03	7.301e−01	SPTA1:33 SLC4A1:312 ANK1:604 EPB42:2260 SPTB:3398 NA
Hyperlipidemia, Familial Combined	0.11397288	62	1.941e−03	7.301e−01	MTTP:36 ALB:51 CYP7A1:131 FADS3:269 TXNIP:363 CUX1:610
Hypoxia−Ischemia, Brain	0.31688639	8	1.913e−03	7.301e−01	NOS2:197 TNF:1163 HIF1A:1560 IRAK1:1736 IL6:1791 VEGFA:1801
PSEUDOHYPOALDOSTERONISM, TYPE IID	0.35584085	6	2.541e−03	7.301e−01	SLC12A3:356 WNK1:975 PRPF4B:1079 TRPV5:1660 WNK4:3177 NEK1:3567
22q11 partial monosomy syndrome	0.28010784	10	2.164e−03	7.301e−01	HIRA:61 COMT:180 ARVCF:608 UFD1:1138 GP1BB:2250 PI4KA:2339
Abnormality of aortic arch	0.27826471	10	2.314e−03	7.301e−01	HIRA:61 COMT:180 ARVCF:608 UFD1:1138 GP1BB:2250 MCTP2:2435
Benign Prostatic Hyperplasia	0.05466704	266	2.304e−03	7.301e−01	CYP11A1:4 GPRC5A:24 PROM1:40 TXNRD3:58 PGC:65 HP:90
Benign recurrent intrahepatic cholestasi	0.37988371	6	1.272e−03	7.301e−01	ABCB11:56 GGTLC3:793 GGT2:1616 ATP8B1:1886 GGT1:2045 GGTLC1:2577
Cardiovascular Diseases	0.03874618	561	1.999e−03	7.301e−01	MTTP:36 ALB:51 TXNRD3:58 PTH:77 HP:90 NPC1:116
Cholestasis	0.06710457	229	5.054e−04	7.301e−01	SLC23A1:11 ABCB11:56 SLC22A1:62 HP:90 NPC1:116 CYP7A1:131
Chronic Kidney Diseases	0.05868537	263	1.138e−03	7.301e−01	ALB:51 SLC22A1:62 PTH:77 HP:90 CLDN16:103 CHDH:127
Classical galactosemia	−0.25887603	14	7.995e−04	7.301e−01	GAL:16 S100A1:21 GALK1:228 LGALS1:259 DMD:324 TGFBI:4699.5
Fatty Liver	0.05037213	312	2.409e−03	7.301e−01	SUGP1:27 MTTP:36 SGSM3:46 ALB:51 HP:90 NPC1:116
Hereditary spherocytosis	0.20908870	21	9.145e−04	7.301e−01	SPTA1:33 SLC4A1:312 ANK1:604 CAD:730 SLC26A8:768 G6PD:828
Hyperemia	0.19775945	25	6.241e−04	7.301e−01	PTH:77 NOS2:197 ADRB1:559 PTGS2:587 HRH1:606 SERPINE1:739
Hypocalcemia	0.13985742	40	2.228e−03	7.301e−01	HIRA:61 PTH:77 CLDN16:103 CHDH:127 COMT:180 SLC4A1:312
Impaired T cell function	0.23614517	15	1.547e−03	7.301e−01	HIRA:61 COMT:180 ARVCF:608 UFD1:1138 TNFRSF13B:1510 SLC39A4:1680
Inborn Errors of Metabolism	0.11224800	61	2.464e−03	7.301e−01	GCDH:78 HEXB:164 AGXT:219 TXNRD1:282 GLA:314 SELENBP1:414
Intestinal Polyps	0.17798546	27	1.377e−03	7.301e−01	CNNM4:204 PLA2G1B:225 NAGLU:349 GUCA2A:558 PTGS2:587 NR4A2:622
Left−Sided Heart Failure	0.09029723	94	2.540e−03	7.301e−01	ALB:51 PTH:77 NOS2:197 NOX1:213 ATP2A1:216 GHRL:252
Liver neoplasms	0.03089985	870	2.531e−03	7.301e−01	SUGP1:27 MTTP:36 PROM1:40 TIGAR:50 ALB:51 ABCB11:56
Nephrolithiasis	0.11583076	76	4.925e−04	7.301e−01	PTH:77 CLDN16:103 CDC73:156 SLC2A9:173 SLC13A2:205 AGXT:219
Persistant truncus arteriosus	0.20322111	19	2.173e−03	7.301e−01	HIRA:61 GJA5:145 COMT:180 ARVCF:608 DLL4:731 MAPK1:882
Recurrent infections	0.13487133	48	1.242e−03	7.301e−01	PRKCD:91 MANBA:104 MC2R:224 SLC46A1:524 CARD11:651 WIPF1:664
Reticulocytosis	0.21924926	20	6.911e−04	7.301e−01	SEC23B:31 SLC4A1:312 KCNN4:547 ANK1:604 G6PD:828 RHAG:855
Sea−Blue Histiocyte Syndrome	0.41437531	5	1.333e−03	7.301e−01	NPC1:116 SMPD1:276 NPC2:467 GLB1:1134 APOE:3350 NA
Seborrheic dermatitis	0.22741693	18	8.401e−04	7.301e−01	HIRA:61 COMT:180 ZNF750:518 ARVCF:608 UFD1:1138 TNF:1163
Spherocytosis	0.39463988	5	2.243e−03	7.301e−01	SPTA1:33 SLC4A1:312 ANK1:604 EPB42:2260 SPTB:3398 NA
Taste sweet	0.25991207	12	1.827e−03	7.301e−01	TAS1R1:385 TAS1R3:521 SLC2A2:899 LPAR2:1039 TAS1R2:1040 ADRA2B:1415
Tetany	0.21303369	19	1.311e−03	7.301e−01	HIRA:61 PTH:77 CLDN16:103 COMT:180 SLC12A3:356 KCNJ1:498
Adenocarcinoma of rectum	0.17318705	24	3.330e−03	7.560e−01	MYD88:125 DCC:159 ENTPD1:166 SLC12A9:271 TYMS:468 TP53BP1:470
Cholangiolocellular Carcinoma	0.41798942	4	3.789e−03	7.560e−01	PROM1:40 ALB:51 CRNKL1:1518 MTHFR:2476 NA NA
Drug−Induced Liver Disease	0.06206984	183	3.947e−03	7.560e−01	IL1R2:17 ALB:51 ABCB11:56 LRAT:112 NR1I2:133 ATR:135
Epidermolysis Bullosa	−0.15632613	30	3.061e−03	7.560e−01	KLHL24:83 KRT14:150 ITGB4:4699.5 TP53:4699.5 LAMC2:4699.5 DST:4699.5
Essential Hypertension	0.05963516	199	3.908e−03	7.560e−01	ALB:51 PTH:77 HP:90 GJA5:145 NOS2:197 NOX1:213
Hyperactive renin−angiotensin system	0.29450108	8	3.924e−03	7.560e−01	SCNN1G:358 SCNN1B:406 KCNJ1:498 SCNN1A:653 NR3C2:1184 SLC26A3:1580

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002136 abnormal kidney physiology	0.07397620	245	9.818e−05	4.673e−02	SLC23A1:11 BCAT2:28 AQP11:35 MAN2A1:42 ALB:51 PRKCD:91
MP0005636 abnormal mineral homeostasis	0.09603105	118	3.660e−04	6.061e−02	CYP11A1:4 ALB:51 CLDN16:103 SLC2A9:173 MC2R:224 SLC4A1:312
MP0009642 abnormal blood homeostasis	0.03579236	1034	5.093e−04	6.061e−02	CLPS:1 CYP11A1:4 SLC23A1:11 ENTPD5:20 BCAT2:28 GYS2:34
MP0009643 abnormal urine homeostasis	0.07349587	200	4.387e−04	6.061e−02	SLC23A1:11 SLC1A1:12 BCAT2:28 MAN2A1:42 ALB:51 SLC5A8:64
MP0005360 urolithiasis	0.34058404	7	1.817e−03	9.909e−02	SLC2A9:173 AGXT:219 SLC3A1:861 OTC:2985 SLC26A6:3075 SLC7A9:3164
MP0003718 maternal effect	0.19406571	23	1.304e−03	9.909e−02	SLC23A1:11 RBP2:57 BUB1:109 PHEX:496 DPPA4:632 PLA2G4A:1110
MP0001663 abnormal digestive system	0.10818516	70	1.873e−03	9.909e−02	SLC22A1:62 SLC4A9:176 GLG1:257 SLC9A4:319 SCNN1B:406 PRKDC:473
MP0002118 abnormal lipid homeostasis	0.05225125	338	1.402e−03	9.909e−02	CLPS:1 CYP11A1:4 ENTPD5:20 BCAT2:28 EHHADH:29 GYS2:34
MP0005464 abnormal platelet physiology	0.11076947	67	1.833e−03	9.909e−02	MERTK:70 ENTPD1:166 NOS2:197 VASP:310 FERMT3:369 LAT:507
MP0001765 abnormal ion homeostasis	0.09247759	92	2.371e−03	1.000e−01	CYP11A1:4 CLDN16:103 SLC2A9:173 MC2R:224 SLC26A4:296 SLC4A1:312
MP0005381 digestive/alimentary phenotyp	0.09236812	92	2.399e−03	1.000e−01	SLC22A1:62 CDC73:156 SLC4A9:176 GLG1:257 SLC9A4:319 MYH11:346
MP0001986 abnormal taste sensitivity	0.29101144	9	2.521e−03	1.000e−01	FFAR1:268 TAS1R1:385 TAS1R3:521 TAS1R2:1040 TRPA1:2660 TRPV1:3187
MP0000609 abnormal liver physiology	0.04787824	322	4.202e−03	1.406e−01	ENTPD5:20 EHHADH:29 GYS2:34 MTTP:36 MAN2A1:42 ALB:51
MP0003656 abnormal erythrocyte physiolo	0.18889992	19	4.431e−03	1.406e−01	MAN2A1:42 SLC4A1:312 KCNN4:547 ANK1:604 PIGA:881 STEAP3:1532
MP0005395 other phenotype	0.09995833	69	4.333e−03	1.406e−01	SLC23A1:11 RBP2:57 BUB1:109 CPT1A:163 ENTPD1:166 AGXT:219
MP0004043 abnormal pH regulation	0.23336240	12	5.172e−03	1.539e−01	SCNN1B:406 KCNJ1:498 RHCG:813 SLC9A3:1117 SOD2:1246 NOS1:1904
MP0000598 abnormal liver morphology	0.03988654	391	9.186e−03	2.495e−01	ENTPD5:20 GYS2:34 MTTP:36 PNLIP:37 MAN2A1:42 ABCB11:56
MP0003868 abnormal feces composition	0.15340777	24	9.435e−03	2.495e−01	CLPS:1 PNLIP:37 ALB:51 CYP7A1:131 RAD23B:503 EIF2AK3:596
MP0000955 abnormal spinal cord	−0.05874959	161	1.130e−02	2.640e−01	WNT4:25 CHUK:38 MOG:87 DISP1:101 PITPNA:162 CHRNA1:164
MP0001666 abnormal nutrient absorption	0.11678371	40	1.088e−02	2.640e−01	CLPS:1 SLC23A1:11 PNLIP:37 ALB:51 RBP2:57 CYP7A1:131
MP0002184 abnormal innervation	−0.07229476	104	1.165e−02	2.640e−01	CHRNA1:164 HOXB13:173 EGLN3:4699.5 PTPN11:4699.5 VAV2:4699.5 NGFR:4699.5
MP0005332 abnormal amino acid	0.07132248	104	1.282e−02	2.774e−01	BCAT2:28 SLC2A9:173 GNMT:174 NOS2:197 ADORA1:370 PTGIS:447
MP0008872 abnormal physiological respon	0.10156644	47	1.646e−02	3.406e−01	NR1I2:133 DRD3:243 SLC4A1:312 SLC12A3:356 ADRB1:559 CYP1A1:695
MP0003091 abnormal cell migration	−0.09173109	57	1.718e−02	3.407e−01	CHUK:38 MAP3K1:172 MAPK8:211 BRCA1:247 KCNJ6:342 ARID5B:397
MP0003453 abnormal keratinocyte physiol	−0.09791973	47	2.075e−02	3.951e−01	PTEN:5 CHUK:38 KRT14:150 BRCA1:247 RAC1:451 STAT3:4699.5
MP0005330 cardiomyopathy	0.08988191	55	2.179e−02	3.989e−01	GJA5:145 MYH11:346 HBEGF:446 HADHB:466 SLC6A6:478 TRIM54:774
MP0002332 abnormal exercise endurance	0.12609480	26	2.640e−02	4.654e−01	GYS2:34 ACSS1:411 SLC6A6:478 PTGS2:587 SLC18A3:985 SOD2:1246
MP0005365 abnormal bile salt	0.11948562	26	3.539e−02	5.614e−01	EHHADH:29 PNLIP:37 ALB:51 ABCB11:56 CYP7A1:131 SLC13A1:798
MP0002078 abnormal glucose homeostasis	0.03328150	368	3.444e−02	5.614e−01	ENTPD5:20 BCAT2:28 GYS2:34 PNLIP:37 ABCB11:56 PRKCD:91
MP0010329 abnormal lipoprotein level	0.07226763	73	3.396e−02	5.614e−01	CLPS:1 MTTP:36 SMPD1:276 TXNIP:363 RORA:413 MOGAT2:672
MP0002135 abnormal kidney morphology	0.03158430	380	4.171e−02	6.204e−01	BCAT2:28 AQP11:35 MAN2A1:42 GCDH:78 PRKCD:91 MANBA:104
MP0004019 abnormal vitamin homeostasis	0.10127052	34	4.163e−02	6.204e−01	SLC23A1:11 RBP2:57 CLDN16:103 LRAT:112 CYP7A1:131 CUBN:226
MP0010678 abnormal skin adnexa	−0.11856924	23	4.949e−02	6.454e−01	PTEN:5 CHUK:38 KRT14:150 FZD6:407 DICER1:4699.5 EGFR:4699.5
MP0002092 abnormal eye morphology	−0.06548245	77	4.856e−02	6.454e−01	CHUK:38 FAT1:112 MAP3K1:172 MAPK8:211 IFT172:316 FUZ:370
MP0004811 abnormal neuron physiology	−0.03945021	222	4.738e−02	6.454e−01	PTEN:5 GAL:16 CASK:26 CHUK:38 ITSN1:132 ROBO1:249
MP0005503 abnormal tendon morphology	0.14962109	14	5.288e−02	6.454e−01	PCOLCE:378 COL14A1:918 FBN1:1012 MKX:1336 TBX5:1613 FMOD:1649
MP0005319 abnormal enzyme/ coenzyme	0.04838452	144	4.802e−02	6.454e−01	SLC1A1:12 ENTPD5:20 NR1I2:133 HPN:160 GNMT:174 NOS2:197
MP0000627 abnormal mammary gland	−0.05646809	101	5.208e−02	6.454e−01	PTEN:5 GAL:16 PTPRF:20 CHUK:38 GPRC6A:61 BRCA1:247
MP0003186 abnormal redox activity	0.08596861	43	5.203e−02	6.454e−01	GPX4:371 NQO1:374 NR4A2:622 CBS:635 GSTZ1:673 XPC:765
MP0002177 abnormal outer ear	−0.06925065	63	5.877e−02	6.994e−01	CHUK:38 PAX6:376 CISD2:425 NCOA3:429 PTPN11:4699.5 COL1A1:4699.5

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
liver	0.081064035	294	2.697e−06	1.456e−04	EHHADH:29 GYS2:34 ALB:51 ABCB11:56 SLC22A1:62 TM4SF4:86
peyers.patch	0.086560413	152	2.638e−04	7.123e−03	SLC23A1:11 MTTP:36 RBP2:57 LGALS4:60 SLC5A12:67 NR1I2:133
cortex.kidney	0.102536758	100	4.294e−04	7.728e−03	SLC5A12:67 ENAM:68 TMEM139:81 CLDN16:103 SLC6A12:119 SLC36A2:138
stomach	0.108535132	74	1.311e−03	1.416e−02	PGC:65 CLDN18:182 TM4SF5:185 EPN3:214 GHRL:252 A4GNT:267
transverse.colon	0.083958913	128	1.137e−03	1.416e−02	LGALS4:60 TSPAN1:108 NR1I2:133 FABP1:179 TM4SF5:185 OMP:199
hypothalamus	−0.103293732	45	1.681e−02	1.296e−01	TMEM196:31 SST:126 SYTL5:186 PCDH15:4699.5 GHRH:4699.5 NPY2R:4699.5
thyroid	0.063271413	126	1.495e−02	1.296e−01	HIRA:61 SLC5A8:64 FHDC1:124 MGAT4C:193 TMEM30B:249 IVD:264
Brodmann.area.9	−0.056427004	122	3.265e−02	2.089e−01	NRCAM:15 DACH2:63 SYT1:114 KCNIP4:241 TMEM178B:314 NEUROD6:4699.5
pituitary.gland	−0.047423445	170	3.481e−02	2.089e−01	GAL:16 TMEM196:31 CRHR2:94 IRS4:148 OTOG:374 DLK1:4699.5
eye.development	−0.090783802	37	5.655e−02	3.054e−01	TKT:231 PAX6:376 FZD6:407 FGFR2:4699.5 OTX2:4699.5 LHX2:4699.5
endocervix	−0.069193998	59	6.699e−02	3.289e−01	TGM2:44 GUCY1A2:127 ISLR:399 VIPR2:453 TRHDE:4699.5 ADGRA2:4699.5
left.ventricle	−0.062974443	57	1.012e−01	3.905e−01	FSD2:18 S100A1:21 HSPB3:76 SMPX:212 NRAP:234 ASB18:290
subcutaneous.adipose	−0.063216157	61	8.889e−02	3.905e−01	KLB:223 THRSP:304 ADIPOQ:4699.5 LEP:4699.5 PLIN1:4699.5 PKD1L2:4699.5
transformed.skin.fibroblast	−0.032828614	221	9.724e−02	3.905e−01	PCDHGA11:53 FAT1:112 NEDD4:179 LRP1:192 ANXA5:193 CRIM1:207
amygdala	−0.135597549	7	2.143e−01	5.259e−01	BCAN:4699.5 LHFPL3:4699.5 NEUROD6:4699.5 PCDH15:4699.5 PTPRZ1:4699.5 SLC38A8:4699.5
coronary.artery	0.104320696	13	1.931e−01	5.259e−01	GJA5:145 CCN3:794 MTHFD1L:2580 PCOLCE2:2876 CCDC190:3068 MYO18B:3151
pancreas	0.040694977	81	2.077e−01	5.259e−01	CLPS:1 PNLIP:37 PRSS3:49 CPB1:74 GNMT:174 CTRC:215
putamen	0.121912792	10	1.821e−01	5.259e−01	RGS9:360 SLC18A3:985 GRM3:1286 DIPK1C:1620 OPALIN:2912 ANO3:8031.5
sigmoid.colon	−0.062022139	39	1.812e−01	5.259e−01	TACR2:72 PHOX2A:449 NKX2−3:4699.5 BMP3:4699.5 BTC:4699.5 CACNA1C:4699.5
skeletal.muscle	−0.028575590	190	1.796e−01	5.259e−01	FSD2:18 HSPB3:76 CHRNA1:164 SMPX:212 NRAP:234 STAC3:306
suprapubic.skin	−0.029198363	174	1.888e−01	5.259e−01	WNT4:25 TFAP2C:124 KRT14:150 CAPNS2:226 CD207:437 LY6D:4699.5
tibial.artery	−0.039709942	85	2.081e−01	5.259e−01	CCN5:390 SPINK13:4699.5 SCUBE3:4699.5 SUSD5:4699.5 ACADL:4699.5 AMIGO2:4699.5
Brodmann.area.24	−0.072606858	21	2.500e−01	5.261e−01	NEUROD6:4699.5 CNTN5:4699.5 CRH:4699.5 DDN:4699.5 DRD5:4699.5 FAM189A1:4699.5
fallopian.tube	−0.052936206	44	2.256e−01	5.261e−01	SERTM1:138 CDON:4699.5 LTC4S:4699.5 CCDC198:4699.5 GASK1A:4699.5 ADAMTS3:4699.5
hippocampus.proper	−0.135574468	6	2.503e−01	5.261e−01	NEUROD6:4699.5 HRK:4699.5 CABP7:4699.5 GRP:4699.5 NEUROG3:4699.5 NTS:4699.5
uterus	−0.048297216	47	2.533e−01	5.261e−01	RERG:215 TCF23:4699.5 TEX15:4699.5 ADAMTS19:4699.5 ATP2B4:4699.5 BNC2:4699.5
breast	−0.093180269	12	2.640e−01	5.281e−01	THRSP:304 ABCB5:4699.5 ADIPOQ:4699.5 ALX4:4699.5 KRT5:4699.5 LEP:4699.5
atrium.auricle	−0.032798022	78	3.189e−01	5.953e−01	FSD2:18 S100A1:21 HSPB3:76 ADAMTS7:153 SMPX:212 NRAP:234
cerebellum	−0.016911416	290	3.307e−01	5.953e−01	GPR179:9 TRIM46:121 FOXD4L5:238 SYT7:262 PAX6:376 CCNP:391
nucleus.accumbens	−0.035394641	64	3.293e−01	5.953e−01	DACH2:63 PCDH11X:69 KCNIP2:201 CSMD3:4699.5 KCNK4:4699.5 ANO3:4699.5
prostate	−0.047567367	30	3.680e−01	6.229e−01	ANO7:151 HOXB13:173 TRPM8:345 CREB3L4:447 KRT5:4699.5 ARHGEF38:4699.5
substantia.nigra	0.082064384	10	3.691e−01	6.229e−01	SPX:581 SLC6A11:646 SLC6A3:845 DBX2:2278 CHRM5:8031.5 FOXB1:8031.5
cerebellar.hemisphere	−0.012773123	363	4.142e−01	6.778e−01	CAMK2B:8 GPR179:9 VAX2:17 STXBP1:64 NBEA:118 CA10:136
leg.skin	−0.016035850	194	4.468e−01	7.096e−01	EREG:103 KRT14:150 ANKRD35:219 CAPNS2:226 MMP27:423 CD207:437
minor.salivary.gland	0.027002755	62	4.638e−01	7.156e−01	BPIFB1:2 PROM1:40 CRACR2B:397 KCNN4:547 SCNN1A:653 ATP6V1B1:705
aorta	−0.016995209	128	5.101e−01	7.248e−01	GABRR1:35 PCDH11X:69 ACTN1:4699.5 ADAMTSL3:4699.5 AEBP1:4699.5 ALCAM:4699.5
blood	0.013281856	235	4.896e−01	7.248e−01	IL1R2:17 HBQ1:53 MYD88:125 TREML2:136 ITGAX:152 VAV1:155
testis	0.006308926	1200	4.994e−01	7.248e−01	ACRBP:7 LYPD4:9 COPRS:21 TMCO5A:32 SPTA1:33 AQP11:35
cerebral.cortex	0.016740899	78	6.109e−01	8.284e−01	COL26A1:165 NOS2:197 DIRAS1:721 NCAN:1018 NPAS1:1115 MAS1:1192
vagina	0.016828732	76	6.136e−01	8.284e−01	CERS3:304 SCNN1G:358 FAT2:430 EDN3:609 NUAK2:682 SULT2B1:692